



Assignment 1

Object Detection for Wildlife Conservation - Detecting
Waterfowl in UAV Thermal Imagery



Automated Waterfowl Detection

Baseline (Thermal-Only Detection)

Data Overview:

- **Source:** UAV-derived Thermal and RGB images (Mendeley Data).
- **Thermal Images:** Low-resolution (512x640), single-channel (grayscale).
- **Objects:** Very challenging due to extremely small size (all bounding boxes are 7x7 pixels).
- **Dataset Size:** 542 total thermal images (355 positive/waterfowl-present).



Training Pipeline

1. Data Preparation

- Convert CSV annotations to YOLO format: (x_center, y_center, w, h)
- Conversion of 1 Channel Thermal to 3 Channel images.
- Organize dataset structure for training

2. Model Selection

- Models: YOLOv8n (nano variant) and YOLOv5n
- Lightweight architecture for edge devices
- Fastest and smallest yolo models
- Load pre-trained weights for transfer learning

3. Data Augmentation

- Mosaic augmentation (combine 4 images)
- HSV color space adjustments
- Random scaling and resizing
- Horizontal/vertical flipping

4. Training Configuration

- Optimizer: AdamW (adaptive learning rate)
- Training epochs: 93 (early stopped)
- Patience: 30 epochs (no improvement threshold)
- Transfer learning from pre-trained weights

5. Model Output

- Trained YOLOv8n model checkpoint
- Optimized for real-time UAV inference
- Ready for edge deployment
- Early stopping prevents overfitting



Evaluation Results(Yolo v8)

THERMAL ONLY MODEL - TRAIN SET RESULTS

mAP50-95: 0.489 (Mean AP across IoU 0.5-0.95)

mAP50: 0.905 (Mean AP at IoU=0.5)

Precision: 0.868 (What % of detections are correct)

Recall: 0.885 (What % of birds are detected)

THERMAL ONLY MODEL - TEST SET RESULTS

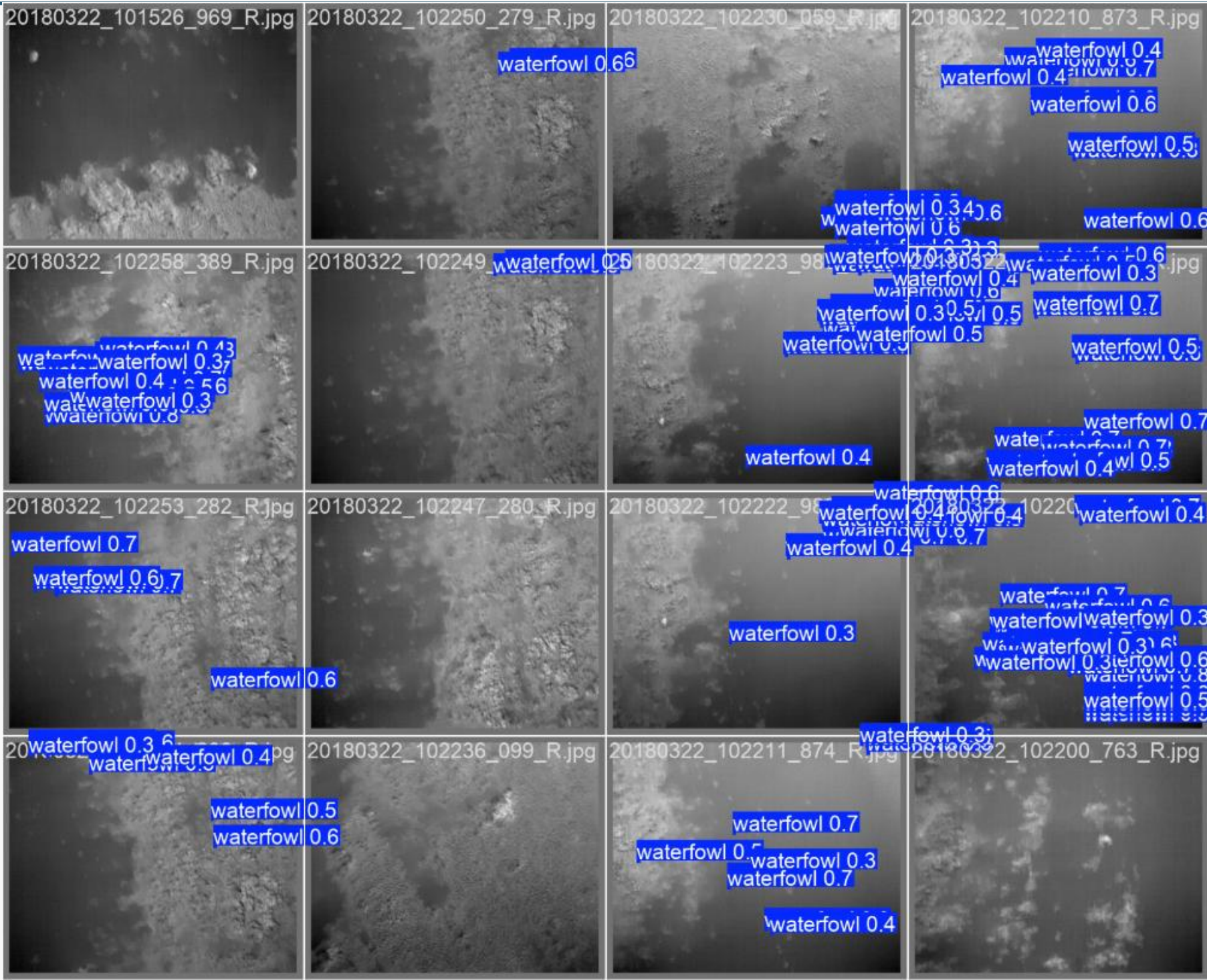
mAP50-95: 0.5037 (Mean AP across IoU 0.5-0.95)

mAP50: 0.8653 (Mean AP at IoU=0.5)

mAP75: 0.5158 (Mean AP at IoU=0.75)

Precision: 0.9257 (What % of detections are correct)

Recall: 0.7957 (What % of birds are detected)



Evaluation Results (Yolo v5)

THERMAL ONLY MODEL - TRAIN SET RESULTS

mAP50-95: 0.595 (Mean AP across IoU 0.5-0.95)

mAP50: 0.956 (Mean AP at IoU=0.5)

Precision: 0.941 (What % of detections are correct)

Recall: 0.937 (What % of birds are detected)

THERMAL ONLY MODEL - TEST SET RESULTS

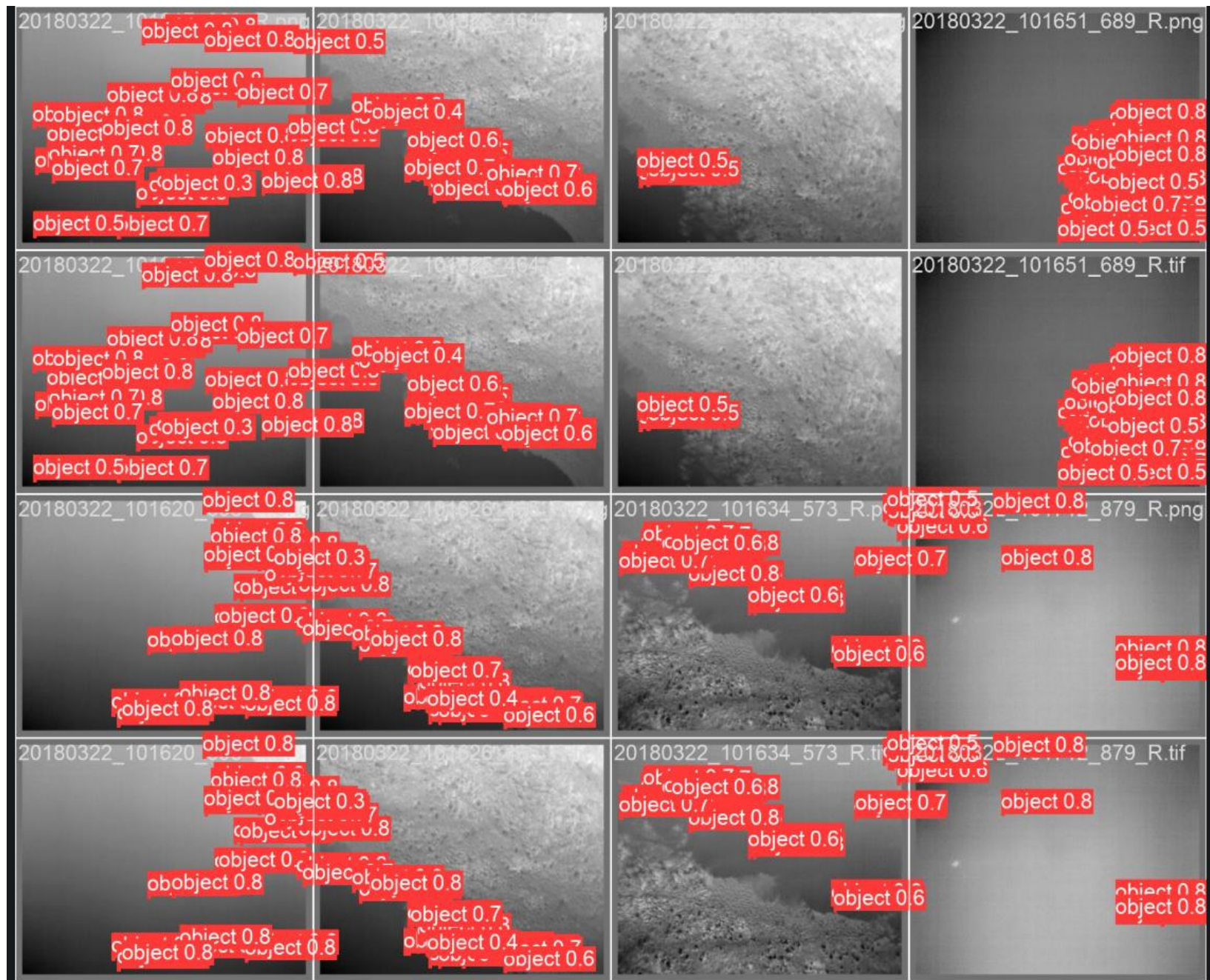
mAP50-95: 0.545 (Mean AP across IoU 0.5-0.95)

mAP50: 0.946 (Mean AP at IoU=0.5)

mAP75: 0.5158 (Mean AP at IoU=0.75)

Precision: 0.941 (What % of detections are correct)

Recall: 0.921 (What % of birds are detected)



Automated Waterfowl Detection

Hypothesized (RGB Fusion Detection)

Data Overview:

- **Source:** UAV-derived Thermal and RGB images (Mendeley Data).
- **Thermal Images:** High-resolution (3000x4000), triple-channel (RGB).
- **Objects:** Dependent on Lighting and can be camouflaged.
- **Dataset Size:** 663 total RGB images (355 positive based on thermal mapping).



Evaluation Results

FUSION MODEL - TRAIN SET RESULTS

mAP50-95: 0.358 (Mean AP across IoU 0.5-0.95)

mAP50: 0.783 (Mean AP at IoU=0.5)

Precision: 0.918 (What % of detections are correct)

Recall: 0.641 (What % of birds are detected)

FUSION MODEL - TEST SET RESULTS

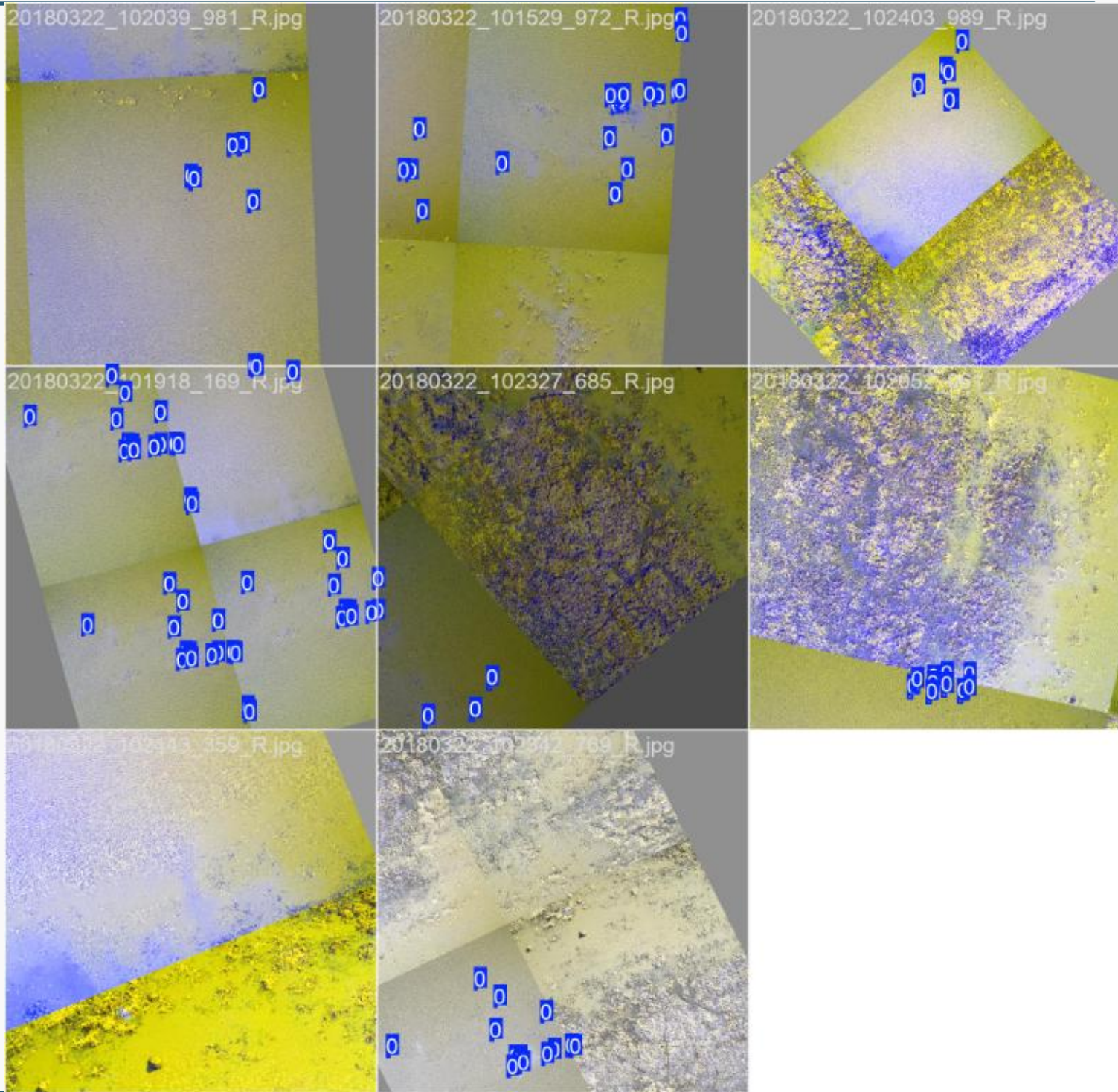
mAP50-95: 0.3577 (Mean AP across IoU 0.5-0.95)

mAP50: 0.7833 (Mean AP at IoU=0.5)

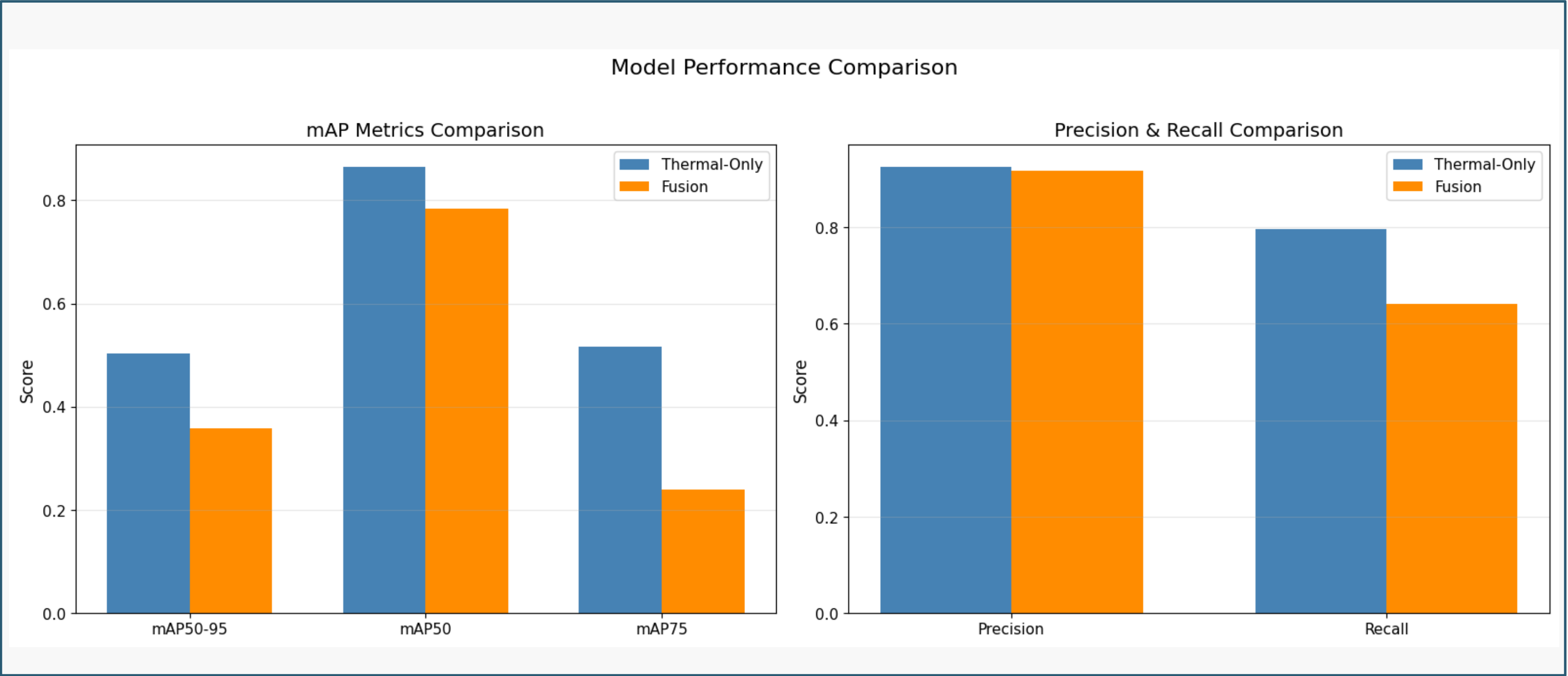
mAP75: 0.2395 (Mean AP at IoU=0.75)

Precision: 0.9178 (What % of detections are correct)

Recall: 0.6415 (What % of birds are detected)



Evaluation Results





Assignment 2

Deep Metric Learning for Waterfowl Detection and
Conservation



Task 1 : Triplet Loss Dynamics

Triplet Loss Function

$$\mathcal{L} = \max(0, d(a,p) - d(a,n) + m),$$

Where,

- Anchor (a) = Reference sample
- Positive (p) = Same class as anchor
- Negative(n) = Different class
- $d(x,y)$ = Euclidean distance between embeddings
- $m > 0$ = margin parameter (minimum required separation)

Goal: Make $d(a,p)$ small and $d(a,n)$ large, with at least margin m between them



Task 1.1: How Margin Affects Inter-Class Distance

Why Does the Margin Matter?

- The margin m acts as a **minimum required separation** in the embedding space. The network is only penalized if the distance difference is greater than m .

The Loss Pushes Until:

$$d(a,n) \geq d(a,p) + m$$

Inter-class distance must be at least m units greater than intra-class distance

Larger Margin m : Forces larger gap between classes → Better separation

Smaller Margin m : Allows classes to be closer together



Task 1.1: Condition for Zero Expected Loss

When Does $\mathbb{E}[\mathcal{L}] = 0$?

The loss becomes zero when: $d(a,p) - d(a,n) + m \leq 0$

Rearranging the terms, the condition for zero loss is:

$$d(a, n) \geq d(a, p) + m$$

Interpretation:

The expected loss becomes zero when, on average across all triplets, the distance between Anchor and Negative is **greater than or equal to** the distance between Anchor and Positive **plus** the margin m .



Task 1.2 Two Failure Modes

Case 1: Margin Too Large – Training Stops

- **Scenario: $\mathcal{L} = 0$ for ALL triplets** : The model just stops learning (Easy Triplets)

- **Why it's Broken:**

If m is set to an extremely high value, the condition $d(a,p) + m > d(a,n)$ is affected

- **The Problem:**

The embedding space is finite and bounded (often by normalization). If margin m is larger than the maximum achievable separation, the zero-loss condition is satisfied by default for virtually all triplets, even before any learning!

- **Consequence:**

Loss $\mathcal{L}$ is consistently zero $\rightarrow$ No gradient signal $\rightarrow$ No learning feedback $\rightarrow$ Weights stop updating $\rightarrow$ Premature halt prevents meaningful convergence

Case 2: Margin Too Small – Cluster Collapse

- **Scenario: Loss approaches 0 too quickly:** Without establishing stable, well-separated clusters

- **Why it's Broken:**

If m is set very close to zero, the requirement $d(a,n) \geq d(a,p) + m$ becomes trivially easy to satisfy

- **The Problem:**

The network expends minimal effort to separate classes. Inter-class distance $d(a,n)$ becomes nearly identical to intra-class distance $d(a,p)$. Without sufficient penalty, the training signal is weak and the embedding space can collapse!

- **Consequence:**

Training converges prematurely without meaningful separation. The embedding space lacks discriminative power, making it ineffective for classification or OOD detection.



Task 2.1: OOD Detection Method

Goal: Detect Non-Waterfowl Samples

OOD = background vegetation, non-bird objects, other wildlife

Why This Method?

We use **Centroid-Based Approach** (Distance-Based Method). It's the easiest way to check how far a new thermal image is from known waterfowl centers.

Step 1: Get the Map Centers

- Calculate Class Prototypes (Centroids) - average embedding for each waterfowl species

Step 2: Calc Weirdness Score

$$S_{OOD}(x) = \min_{c \in C} \{d(E(x), \mu_c)\}$$

The closest distance to any center

Step 3: Make the Call

- Set Threshold τ

$$S_{OOD}(x) > \tau$$

$\Rightarrow$ It's Out – of – Distribution!



Task 2.2: Mathematical Proof

Goal:

Formally prove that if clusters are nice and tight, OOD samples will have larger minimum distance than ID samples

Ideal Assumptions:

- **Tight Clusters:** Each class c sits in a hypersphere of radius R_c .
- **ID is Inside:** Any ID sample has distance $\leq R_c$ to its center
- **OOD is Outside:** Any OOD sample is outside every cluster

The Proof:

1. **ID Samples:** Minimum distance $\text{Dist}(x_{ID}) \leq R_c$ (distance to own center)
2. **OOD Samples:** Distance to every center $> R_c$. (outside all clusters)
3. Let R_{max} = radius of biggest cluster

The Key Inequality:

$$S_{OOD}(x_{OOD}) > R_{max} \geq S_{OOD}(x_{ID})$$

This proves: if training worked well (tight clusters), minimum distance for OOD must be higher than for ID!





Thank you

